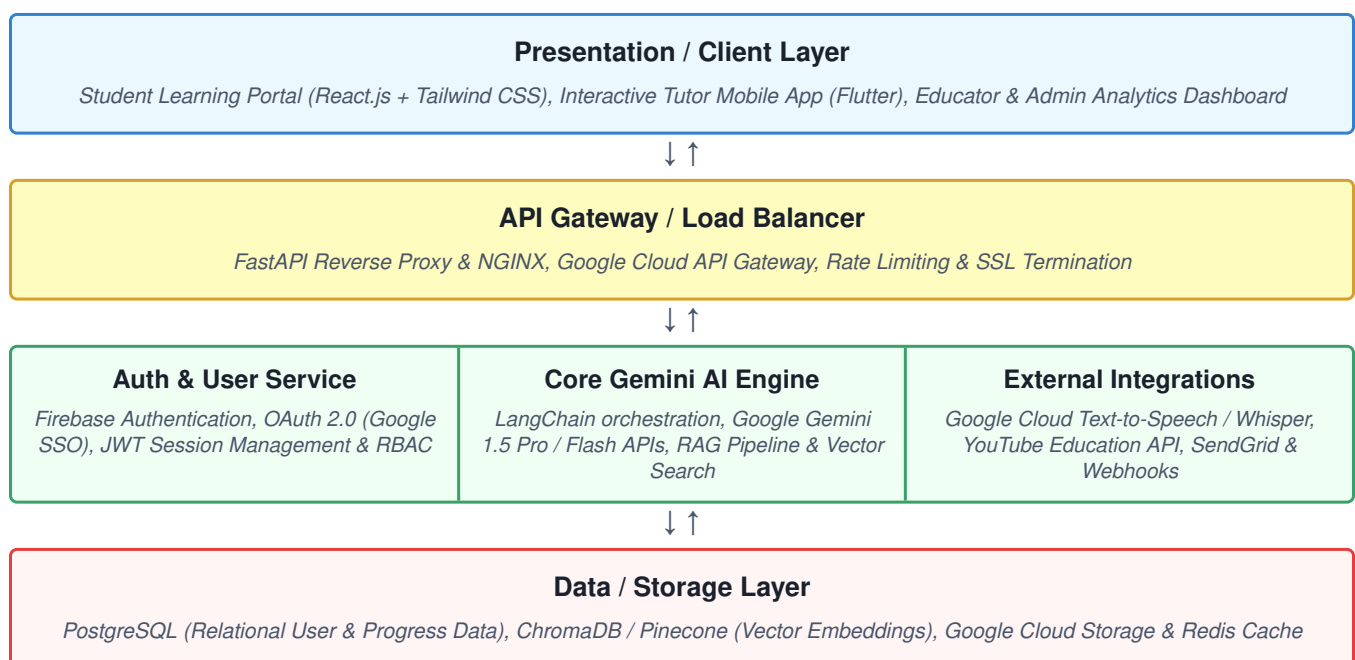


Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-1156
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer (Client)	Provides responsive web and mobile interfaces for students and teachers. Enables interactive AI tutoring, real-time chat, automated quiz generation, personalized learning roadmaps, and student progress tracking dashboards.	React.jsFlutterTailwind CSSViteChart.js
API Gateway / Load Balancer	Acts as the single entry point for all frontend requests. Manages traffic distribution, SSL/TLS offloading, CORS policies, request validation, authentication token verification, and dynamic rate limiting to protect LLM endpoints.	NGINXGoogle Cloud API GatewayFastAPI Reverse Proxy
Authentication Service	Handles secure student and educator onboarding, passwordless Google SSO, session token generation (JWT), and role-based access control (RBAC) to ensure student data privacy and system compliance.	Firebase AuthGoogle OAuth 2.0PyJWTbcrypt

Component Name	Description / Role in Architecture	Technologies Used
Core AI & Logic Service	The primary intelligence module orchestrating context retrieval (RAG) over uploaded course syllabi/notes, formulating customized prompts, querying Gemini models for step-by-step tutoring, and grading student assessments.	Google Gemini 1.5 Pro/Flash LangChain Python FastAPI Sentence-Transformers
External Services & Integrations	Enriches learning with multimodal features: speech-to-text for spoken voice queries, natural speech output for interactive read-aloud, automated notification delivery, and curating relevant educational video links.	Google Cloud TTS Whisper API YouTube Data API v3 SendGrid
Data & Storage Layer	Manages persistence: structured databases for student accounts, quiz histories, and course records; high-performance vector databases for syllabus embeddings; and object storage for study materials and PDFs.	PostgreSQL ChromaDB / Pinecone Google Cloud Storage (GCS) Redis